

TASK-4: E-COMMERCE DASHBOARD

Power BI Project Report

1. Project Title

E-commerce Sales and Performance Dashboard using Power BI

2. Project Objective

The objective of this project is to develop an interactive E-commerce Dashboard using Microsoft Power BI.

The dashboard analyzes sales, revenue, profit, customers, products, and order performance. It provides users with interactive filtering and drill-through functionality to analyze individual products and customers.

The project includes:

- Loading and preparing order and product data
 - Creating revenue and profit measures
 - Calculating return rate
 - Creating monthly sales analysis
 - Identifying top-performing products
 - Analyzing customer distribution
 - Tracking profit trends
 - Adding interactive slicers
 - Creating product and customer drill-through pages
 - Publishing the Power BI report
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3. Dataset

The project is based on the **Superstore Retail Analytics** dataset.

Dataset Source

GitHub repository:

<https://github.com/virajbhutada/bi-projects-collection/tree/main/Superstore%20Retail%20Analytics>

The cleaned dataset used in the Power BI report is named:

cleaned_ecommerce

The dataset contains information related to orders, customers, products, sales, quantities, categories, regions, and profit.

4. Data Preparation

The dataset was imported into Power BI and cleaned using Power Query.

The cleaned data was loaded into a table named:

cleaned_ecommerce

The data preparation process included:

- Removing unnecessary columns
 - Checking missing values
 - Correcting data types
 - Formatting date fields
 - Preparing sales and profit fields for analysis
 - Preparing product and customer information
 - Checking duplicate and invalid records
 - Preparing the data for Power BI relationships and visualizations
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5. Data Model

A separate Date table was created to support time-based analysis.

The Date table contains:

- Date
- Year
- Month
- Month Number
- Year Month
- Year Month Number

The relationship between the Date table and the cleaned ecommerce table is:

DateTable[Date] → cleaned_ecommerce[Order Date]

The relationship is configured as a one-to-many relationship.

This Date table is used for monthly sales and profit trend analysis.

6. DAX Measures

The following measures were created for the dashboard.

Total Revenue

Total Revenue =

SUM(cleaned_ecommerce[Sales])

Total Profit

Total Profit =

SUM(cleaned_ecommerce[Profit])

Total Quantity

Total Quantity =

SUM(cleaned_ecommerce[Quantity])

Total Orders

The order-count measure uses the available order identifier in the cleaned dataset.

Total Orders =

DISTINCTCOUNT(cleaned_ecommerce[Order ID])

Total Customers

Total Customers =

DISTINCTCOUNT(cleaned_ecommerce[Customer ID])

Profit Margin

Profit Margin =

DIVIDE(

[Total Profit],

[Total Revenue],

```

0
)

Return Rate

Where the cleaned dataset contains a Returns field:

Returned Orders =

CALCULATE(

    DISTINCTCOUNT(cleaned_ecommerce[Order ID]),

    cleaned_ecommerce[Returns] = "Yes"

)

Return Rate =

DIVIDE(

    [Returned Orders],

    [Total Orders],

    0

)

```

7. Dashboard Pages

The Power BI report contains three main pages.

Page 1 — E-commerce Dashboard

The main dashboard provides an overall view of ecommerce performance.

It contains KPI cards for:

- Total Revenue
- Total Profit
- Total Orders
- Total Customers
- Return Rate

It also contains:

- Monthly Sales Trend

- Top 10 Products
 - Customer Distribution
 - Profit Trend
 - Interactive slicers
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8. Monthly Sales Analysis

A line chart was created to analyze revenue over time.

Chart fields

X-axis:

DateTable[Year Month]

Y-axis:

[Total Revenue]

The chart allows users to identify changes in sales performance across different months and years.

9. Top Product Analysis

A bar chart was created to identify the top-performing products.

The product visual uses:

Product Name

and

Total Revenue

A Top N filter is applied to display the top 10 products.

This visual helps identify products that contribute the highest revenue.

10. Customer Distribution

A customer distribution visual was created using customer segments.

The visual compares the number of customers across different segments.

Example segments include:

- Consumer
- Corporate
- Home Office

This allows users to understand the distribution of customers across different business segments.

11. Profit Trend

A line chart was created to analyze profit over time.

Chart fields

X-axis:

DateTable[Year Month]

Y-axis:

[Total Profit]

The chart helps identify periods of increasing or decreasing profitability.

12. Interactive Slicers

Interactive slicers were added to allow users to dynamically filter the dashboard.

The main slicers include:

- Year
- Region
- Category
- Segment

When a slicer is selected, the KPI cards and charts update automatically.

This makes the dashboard interactive and allows users to analyze specific portions of the dataset.

13. Product Details Drill-through

A separate **Product Details** page was created using Power BI drill-through functionality.

The drill-through field is:

Product Name

The page contains:

- Revenue
- Profit
- Orders
- Quantity
- Product Sales Trend
- Order Details table
- Back button

The Order Details table is designed to display relevant order-level information such as:

- Order ID
- Order Date
- Customer
- Sales
- Quantity
- Profit
- Returns, where available

Users can right-click a product from the main dashboard and select:

Drill through → Product Details

The page then displays information specific to the selected product.

14. Customer Details Drill-through

A separate **Customer Details** page was created using Power BI drill-through functionality.

The drill-through field is:

Customer Name

The page contains:

- Revenue
- Profit

- Orders
- Quantity
- Customer Sales Trend
- Purchase Details table
- Back button

The Purchase Details table contains relevant fields such as:

- Product
- Category
- Sales
- Quantity
- Profit
- Returns, where available

Users can select a customer and drill through to view detailed purchasing information.

15. Dashboard Design

The dashboard uses a simple and professional layout.

The main design includes:

- KPI cards at the top
- Sales trend in the center
- Product and customer analysis below
- Profit trend toward the bottom
- Slicers for interactive filtering

A consistent color scheme was used to improve readability and make important metrics easy to identify.

16. Key Features

The completed Power BI report provides the following functionality:

Feature	Status
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Feature	Status
Data loading	Completed
Data cleaning	Completed
Revenue calculation	Completed
Profit calculation	Completed
Return rate	Completed where Returns data is available
Monthly sales chart	Completed
Top product analysis	Completed
Customer distribution	Completed
Profit trend	Completed
Interactive slicers	Completed
Product drill-through	Completed
Customer drill-through	Completed
Power BI publishing	Completed

17. Business Insights

The dashboard can be used to identify:

1. Overall revenue and profit performance.
2. Monthly changes in sales.
3. Products generating the highest revenue.
4. Customer segment distribution.
5. Periods with higher or lower profitability.
6. Customer-level purchasing behavior.
7. Product-level sales performance.
8. Returned orders and return rate where return information is available.
9. Regional and category-level performance through slicers.

These insights can help businesses understand sales performance and identify areas requiring improvement.

18. Power BI Publishing

After completing the dashboard, the Power BI report was saved as:

Ecommerce_Dashboard.pbix

The report was published to Power BI Service.

The published report can be shared with authorized users through the Power BI workspace or report link, depending on the organization's sharing permissions.

19. Project Deliverables

The final project consists of:

1. Cleaned Dataset

cleaned_ecommerce.csv

2. Power BI Report

Ecommerce_Dashboard.pbix

3. Dashboard Pages

- E-commerce Dashboard
- Product Details
- Customer Details

4. Screenshots

- Main Dashboard
- Product Details
- Customer Details
- Data Model
- Published Power BI Report

5. Project Report

Project_Report.pdf

20. Conclusion

The E-commerce Dashboard successfully demonstrates the use of Microsoft Power BI for ecommerce data analysis.

The report combines data preparation, DAX calculations, interactive visualizations, slicers, and drill-through functionality into a single analytical solution.

The dashboard provides an easy way to monitor revenue, profit, orders, customers, products, and sales trends while allowing users to explore individual products and customers in greater detail.

This project demonstrates practical skills in data cleaning, data modeling, DAX, visualization, dashboard design, interactive reporting, and Power BI publishing.